Final exam review

A time-series model:

more similar

$y_t = \beta_0 + \beta_1 x_t + u_t$ ← Static: Ignores time, dynamic, ... all effects are immediate (contemporaneous)

$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + u_t$ ← Dynamic with lagged expl. vars.

$y_t = \beta_0 + \beta_1 x_t + \beta_2 x_{t-1} + \beta_3 y_{t-1} + u_t$ ← Dynamic with lagged outcome and expl. vars (ADL)

Two types of exogeneity:

(1) "Contemporaneous" exog: the regressors and dist. are ind. in the given period

(2) "Strict" exogeneity: the regressors are ind. of the dist. in all periods. (includes "contemp" and "cross-period" exog.)

OLS's Unbiasedness for β's requires "strict" exog.

↳ Models w/. lagged outcomes violate strict exog. (the cross-period part).

↳ Models w/out lagged outcomes can still be unbiased.*

OLS's consistency for β's relies on contemporaneous exog.

↳ Models w/. lagged outcomes can be est. consistently IF the dist. is not autocorrelated.

↳ Models w/out lagged outcomes are good.*

*: Autocorrelation causes inference problems and efficiency issues similar to heterosk. & corr. dist.
Solutions: Check specification, NW standard errors, and/or FGLS.

Autocorrelation:

$u_t = \rho u_{t-1} + \varepsilon_t$: AR(1)

$u_t = \rho_1 u_{t-1} + \rho_2 u_{t-2} + \rho_3 u_{t-3} + \varepsilon_t$: AR(3)

"ρ" gives the amt & direction of autocorrelation ($-1 < \rho < 1$)

Correlation ⇒ violated cont. exog.

$y_t = \beta_0 + \beta_1 x_t + \beta_2 y_{t-1} + u_t$

$u_t = \rho u_{t-1} + \varepsilon_t$

$y_{t-1} = \beta_0 + \beta_1 x_{t-1} + \beta_2 y_{t-2} + u_{t-1}$

	Static	Dyn. w lagged exp.	Dyn. w/. lagged outcome
Cont. exog. Consistent?	Y	Y	Yes if no autocorrelation
Unbiased? Strict exog.	Y	Y	Never

(Non-) Stationarity

Goal of stationarity: We need well-behaved "variables" to be able to learn anything.

1. Mean: $\mathbb{E}[X_t] = \mathbb{E}[X_s]$ for $t \neq s$ (all $t \neq s$): Mean of var. is ind. of time.
2. Var: $Var(X_t) = Var(X_s)$ for $t \neq s$: Var. of our var. is also ind. of time. (Violated by random walks)
3. Cov: $Cov(X_t, X_{t-k}) = Cov(X_s, X_{s-k})$ for all $t \neq s$.

Classic ex. of non-stationarity: Random walk:

$X_t = X_{t-1} + \varepsilon_t \Rightarrow$ $Var(X)$ grows w/. time.

Note: A lot like AR(1) w/. $\rho = 1$...

Classic solution: Differencing: $\Delta X_t = X_t - X_{t-1}$

$$= \underbrace{(X_{t-1} + \varepsilon_t)}_{= X_t} - X_{t-1}$$

$$= \cancel{X_{t-1}} + \varepsilon_t - \cancel{X_{t-1}}$$

$$= \varepsilon_t$$ which is stationary. :)

Models w/. non-stationary data

whereas $y_t = \beta_0 + \beta_1 X_t + u_t$ is prone to spurious results when y & X are nonstationary

finding something that's not really there

a Solution: differences

$$\left.\begin{aligned} y_t &= \cancel{\beta_0} + \beta_1 X_t + u_t \\ y_{t-1} &= \cancel{\beta_0} + \beta_1 X_{t-1} + u_{t-1} \end{aligned}\right\}$$ now difference...

$$\underbrace{y_t - y_{t-1}}_{= \Delta y_t} = \underbrace{\beta_1 X_t - \beta_1 X_{t-1}}_{= \beta_1 \Delta X_t} + \underbrace{u_t - u_{t-1}}_{= \Delta u_t}$$

$$\Delta y_t = \beta_1 \Delta X_t + \Delta u_t$$

This model is less likely to provide spurious results and still estimates the same parameter.

Notation:

y: outcome

x: expl. var.

u: disturbance (sometimes ε)

pop. reg.

$y = \beta_0 + \beta_1 x + u$

β: coefficient

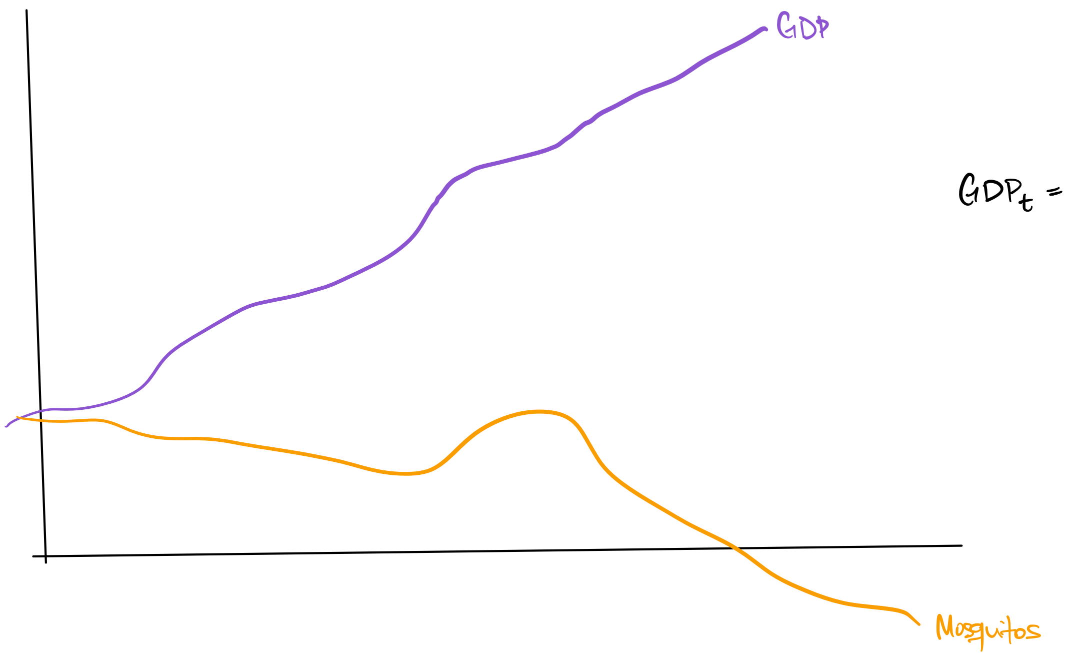
e: residual $= y - \underline{\hat{y}} = y - (\underline{\hat{\beta}_0 + \hat{\beta}_1 x})$

$\hat{\beta}$: estimated coefficient

sample reg. based

$y = \alpha + \beta x + \varepsilon$

$y = \beta_0 + \beta_1 x + u$



$GDP_t = \beta_0 + \beta_1 Mosquitos_t + u_t$